

Capstone Project

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Problem Statement

Build a predictive model to assess heart disease risk using health conditions and behavioral factors.

Summary

This project aims to create a model that predicts heart disease risk by looking at a person's physical health, mental well-being, and lifestyle choices. By using real-world health data and advanced machine learning techniques, we can provide more accurate and personalized risk assessments. The goal is to help people take proactive steps toward heart health while supporting doctors and insurance providers in making informed decisions.

Dataset : <https://www.kaggle.com/datasets/kamilpytlak/personal-key-indicators-of-heart-disease?resource=download>

Data Dictionary

Column Name	Data Type	Description
Heart Disease	Char	Whether the individual has heart disease (Yes/No).
BMI	Float	Body Mass Index (BMI) of the individual.
Smoking	Char	Whether the individual has smoked at least 100 cigarettes in their lifetime (Yes/No).
Alcohol Drinking	Char	Whether the individual is a heavy drinker (Yes/No).
Stroke	Char	Whether the individual has had a stroke (Yes/No).
Physical Health	Integer	Number of days in the past 30 days the individual had poor physical health.
Mental Health	Integer	Number of days in the past 30 days the individual had poor mental health.
Diff Walking	Char	Whether the individual has difficulty walking or climbing stairs (Yes/No).
Sex	Char	Gender of the individual (Male/Female).
Age Category	Char	Age group of the individual (e.g., 55-59, 80 or older, etc.).
Race	Char	Race of the individual (e.g., White, Black, etc.).
Diabetic	Char	Whether the individual has diabetes (Yes/No/Borderline).
Physical Activity	Char	Whether the individual engages in physical activity other than work (Yes/No).
Gen Health	Char	Self-reported general health status (Excellent, Very good, Good, Fair, Poor).
Sleep Time	Integer	Average number of hours of sleep per day.
Asthma	Char	Whether the individual has asthma (Yes/No).
Kidney Disease	Char	Whether the individual has kidney disease (Yes/No).

Process Flow

1. **Data Collection & Understanding** – Collect data and explore significant variables explaining health status and lifestyle factors and Analyse how these relate to heart disease risk by identifying patterns in the data.
2. **Data Preprocessing** – Clean the data by converting medical history and lifestyle factors into structured formats. Handle missing or inconsistent data and standardize measures like BMI and sleep duration.
3. **Exploratory Data Analysis (EDA)** – Explore the data to uncover patterns, such as how stress, alcohol, and mobility issues impact heart disease risk. Use visualizations and statistical methods to find correlations.
4. **Feature Selection** – Identify key risk factors like high blood pressure, poor health, and diabetes. Remove less relevant variables to focus on the most important predictors.
5. **Model Development** – Train machine learning models, starting with Logistic Regression. Test more complex models like Random Forest and XGBoost to improve predictions and refine the model.
6. **Model Evaluation & Optimization** – Evaluate models using metrics like precision, recall, and ROC-AUC. Optimize by adjusting hyperparameters and experimenting with feature combinations.
7. **Insights & Interpretation** – Provide insights based on the model, highlighting factors like sleep, stress, and exercise that increase heart disease risk. Offer recommendations for lifestyle changes.
8. **Conclusion & Future Work** – Summarize findings and assess the model's real-world use. Suggest improvements like adding wearable data, enhancing interpretability, or exploring deep learning for more personalized predictions.

Literature Review

Machine Learning Approaches for Heart Disease Prediction

Heart disease remains a leading cause of mortality worldwide, necessitating advancements in predictive modelling to facilitate early detection and intervention. Traditional diagnostic methods often involve subjective assessments, making machine learning (ML) an essential tool in automating and improving prediction accuracy. This review explores various ML-based approaches and their effectiveness in predicting heart disease.

Machine Learning Algorithms in Heart Disease Prediction

Several studies have utilized ML techniques to predict heart disease risk. Shreekanth (2023) implemented Logistic Regression, Random Forest, Decision Trees, and XGBoost to classify heart disease presence with high accuracy ([GitHub](#)). Similarly, Kumar (2023) developed a heart disease prediction system utilizing Naïve Bayes and Decision Tree algorithms, aimed at assisting healthcare professionals in risk assessment ([GitHub](#)).

A comparative study by Ingole et al. (2024) evaluated seven classifiers, including Logistic Regression, Random Forest, Decision Tree, Naïve Bayes, k-Nearest Neighbours, Neural Networks, and Support Vector Machine (SVM). The results indicated that SVM outperformed other models, achieving an accuracy of 91.51% ([arXiv](#)).

Ensemble and Deep Learning Approaches

Ensemble learning techniques have proven effective in heart disease prediction. Ramineni et al. (2024) introduced a stacked ensemble classifier integrating Extra Trees, Random Forest, and XGBoost, achieving an impressive accuracy of 92.34% ([arXiv](#)). Shrivastava (2023) leveraged Random Forest for classification, emphasizing the importance of feature selection in improving model performance ([GitHub](#)).

Furthermore, Suhas (2023) implemented artificial neural networks (ANNs) to model complex relationships in heart disease datasets. ANNs demonstrated strong predictive power, particularly in cases with nonlinear interactions among risk factors ([GitHub](#)).

Feature Selection and Data Preprocessing

Feature selection and data preprocessing play a crucial role in optimizing ML models. Researchers have utilized Chi-square tests and correlation analysis to identify the most relevant features influencing heart disease outcomes. Divya et al. (2024) emphasized the need for data normalization and outlier detection techniques to enhance model robustness, particularly when integrating Internet of Things (IoT) data for continuous monitoring ([Oncology Radiotherapy](#)).

Challenges and Future Directions

Despite the success of ML models in heart disease prediction, several challenges remain. Data quality and availability are significant concerns, as biased or incomplete datasets can limit model generalizability. Interpretability is another critical factor, particularly in deep learning models, where decision-making processes are often opaque. Future research should focus on integrating ML models with clinical decision support systems to improve real-world applicability.

Conclusion

The reviewed studies highlight the effectiveness of ML techniques in predicting heart disease, with ensemble and deep learning methods offering promising results. Further advancements in data processing, feature selection, and model interpretability will enhance the clinical utility of these models, ultimately contributing to improved patient outcomes and early disease detection.

References

1. Shreekant, G. (2023). Heart Disease Prediction Using Machine Learning. [GitHub](#).
2. Kumar, L. (2023). Heart Disease Prediction System. [GitHub](#).
3. Ingole, B. S., Ramineni, V., Bangad, N., Ganeeb, K. K., & Patel, P. (2024). Advancements in Heart Disease Prediction: A Machine Learning Approach for Early Detection and Risk Assessment. [arXiv](#).

4. Ramineni, V., et al. (2024). Ensemble Framework for Cardiovascular Disease Prediction. [arXiv](#).
5. Shrivastava, V. (2023). Heart Disease Classification using Random Forest. [GitHub](#).
6. Suhas. (2023). Heart disease Predictor using Artificial Neural Networks. [GitHub](#).
7. Divya, N., Riyazuddin, M., Ahad, A., Vulapula, S. R., Manjula, A., & Sirajuddin, M. (2024). Predicting Heart Disease Using Machine Learning and IoT Techniques. [Oncology Radiotherapy](#).